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Validation of Pefloxacin for detection of fluoroquinolone (FQ) resistance among *Salmonella* Typhi with special reference to GyrB mutations

Shakila Banu Inayath^{1,2}, Shobha Broor¹, Ruchi Gupta², Priti Agarwal¹, Subhradeep Majumder², Anupam Kumar Anveshi² and Rajni Gaiind^{2,*}

Abstract

Introduction. Fluoroquinolone (FQ) resistant *Salmonella* are classified as high priority pathogens by WHO. FQ resistance among *Salmonella* Typhi has emerged rapidly and is predominantly mediated by mutations in the topoisomerase genes *gyrA*, and *parC*. Mutations in GyrA result in classical FQ resistance (DCS-NAR) i.e. decreased susceptibility to ciprofloxacin (MIC of 0.12 to 0.5 µg ml⁻¹) (DCS) and resistance to nalidixic acid (NAR). Previously a nalidixic acid disc test was proposed for detection of DCS. Recently isolates with non-classical FQ resistance caused by plasmid-mediated quinolone resistance (PMQR) and mutations in GyrB have emerged. These mechanisms also result in DCS but are nalidixic acid susceptible (NAS) and thus pose diagnostic challenges. CLSI and EUCAST have recommended use of 5 µg pefloxacin discs for detection of DCS in *Salmonella*.

Hypothesis. The CLSI and EUCAST recommendations for use of 5 µg pefloxacin for detection of DCS has not been validated on typhoidal *Salmonella* and resistance mediated by GyrB mutation in *Salmonella* species.

Aim. The aim of the present study was to validate the performance of the 5 µg pefloxacin discs to detect isolates of *S. Typhi* with DCS with special reference to GyrB mutations.

Methodology. A total of 180 clinical isolates of *Salmonella* Typhi (2005–2014) were investigated for genetic mechanisms of resistance. Zone diameters for nalidixic acid (30µg), ciprofloxacin (5µg) and pefloxacin (5µg) and minimum inhibitory concentration (MIC) for ciprofloxacin were determined using CLSI guidelines. Performance of the three discs was evaluated to detect FQ resistance in *S. Typhi*.

Results. Topoisomerase mutations in GyrB +/- ParC and GyrB were detected in 112 and 34 isolates respectively. Different mutations have a varied effect on the MIC for ciprofloxacin. The current breakpoints for susceptible (≤0.06 µg ml⁻¹) and non-susceptible (≥0.125 µg ml⁻¹), failed to detect all isolates with a resistance mechanism. Performance of both ciprofloxacin and pefloxacin discs were excellent compared to nalidixic acid in differentiating isolates with non-classical resistance mediated by GyrB from wild-type.

Conclusion. The pefloxacin disc can be used to detect FQ resistance among *S. Typhi*. This is the first report of validation of pefloxacin for detection of FQ resistance in *S. Typhi* mediated by GyrB mutation.

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Keywords: Non-classical fluoroquinolone resistance; quinolone resistance determining region (QRDR); GyrB mutation; *Salmonella* Typhi; Pefloxacin; Decreased ciprofloxacin susceptibility (DCS).

Abbreviations: CA, Categorical agreement; CIP, Ciprofloxacin; CLSI, Clinical and laboratory standards institute; DCS, Decreased Ciprofloxacin Susceptibility; EUCAST, European Committee on Antimicrobial Susceptibility Testing; FQs, Fluoroquinolones; I, Intermediate; ME, Major error; MICs, Minimum inhibitory concentrations; NAR, Nalidixic acid Resistant; NAS, Nalidixic acid susceptible; PCR, Polymerase chain reaction; PMQR, Plasmid mediated quinolone resistance; QRDRs, Quinolone resistance determining regions; R, Resistant; S, Susceptible; VME, Very major error.

INTRODUCTION

Salmonella enterica subsp. *enterica* contribute to a major disease burden worldwide. Diseases caused by *Salmonella* in humans are gastroenteritis or non-typhoidal salmonellosis (*Salmonella* Typhimurium or *Salmonella* Enteritidis) and enteric fever (*Salmonella* Typhi or Paratyphi). Antimicrobials are essential for timely and appropriate clinical management of invasive *Salmonella* infections [1–3].

Fluoroquinolones (FQs) (e.g. ciprofloxacin) are highly efficient against fully susceptible *Salmonella* bacteria (i.e. isolates without any acquired resistance mechanisms), however there is concern about the rapidly increasing fluoroquinolone resistance in *Salmonella* [4–7].

FQ are critically important antimicrobials for use in humans and are at relatively high risk of selection of bacterial resistance. In 2019, WHO AWaRe Classification Database was developed to guide selection and use of essential medicines. Ciprofloxacin has been classified as WATCH group by WHO and hence monitoring of resistance should be prioritized in key pathogens [8]. Moreover, FQ resistant *Salmonella* have been classified as high priority pathogens in 2017 [9]. FQ resistance in typhoidal *Salmonella* is rapidly emerging in the last decade [10]. In view of these recent developments, monitoring of FQ resistance in *Salmonella enterica* using reliable *in vitro* antimicrobial susceptibility tests are critical to addressing of antibiotic-resistant *Salmonella* infection.

Fluoroquinolone resistance in *Salmonella* Typhi is mainly caused by chromosomal mutations in the quinolone resistance-determining regions (QRDRs) of the 'topoisomerase genes, *gyrA* and *parC*' [11–13]. These mutations usually confer stepwise resistance [11]; a single mutation is associated with decreased ciprofloxacin susceptibility (DCS) i.e. MIC of 0.12 to 0.5 µg ml⁻¹, and resistance to nalidixic acid (MIC, 128–512 µg ml⁻¹), whereas two or more mutations result in resistance to ciprofloxacin, MIC ≥1 µg ml⁻¹. Recently other mechanisms of resistance, among *Salmonella*, that also result in DCS but with only a modest increase in MIC of nalidixic acid (MIC 2–16 µg ml⁻¹) have been reported [10, 14, 15]. These mechanisms included plasmid-mediated quinolone resistance (PMQR) mechanisms mediated by *qnr* variants, *aac(6′)-Ib-cr*, *qepA*, and *oqxAB* and mutations in topoisomerase gene *gyrB*. Since infections with *S. Typhi* strains with DCS have been associated with treatment failure with ciprofloxacin and increased disease severity [16, 17], it is important that these isolates are detected during routine antimicrobial susceptibility testing [18]. Isolates with NAS–DCS phenotype are now being increasingly reported among *Salmonella*, however the nalidixic acid disc test fails to detect these isolates of *S. Typhi* [10] and non-typhoidal *Salmonella* with DCS [19]. In view of these developments and to improve the detection of isolates with acquired resistance both, clinical and laboratory standards institute (CLSI) and European Committee on Antimicrobial Susceptibility Testing (EUCAST), introduced species-specific minimum inhibitor concentration (MIC) breakpoints for *Salmonella* and fluoroquinolones [20, 21].

For diagnostic laboratories in resource limited countries, MIC testing is not a feasible option for detection of any resistance, there is an urgent need for a disc test for detection of isolates with DCS irrespective of resistance mechanism. More recently both CLSI and EUCAST have recommended use of 5 µg pefloxacin disc for detection of DCS in *Salmonella* [20, 21].

In our previous report from India [10], we have reported increasing isolation of *S. Typhi* with decreased susceptibility to ciprofloxacin but with susceptibility to nalidixic with nonsynonymous mutation at codon 464 of the *GyrB*. The aim of the present study was to validate the performance of 5 µg pefloxacin disc to detect isolates of *S. Typhi* with altered susceptibility to ciprofloxacin irrespective of the resistance mechanism. Till date no investigation has been undertaken to validate performance of 5 µg pefloxacin disc for detection of DCS mediated by mutations in *gyrB* gene among *Salmonella*.

METHODS

Study isolates

A total of 180 clinical non-duplicate isolates of *Salmonella* Typhi from patients with enteric fever (2005–2014) were included in the study. The isolates were stored at –80 °C and were selected randomly based on MICs of nalidixic acid and ciprofloxacin. This study was conducted in VMMC and Safdarjung Hospital, New Delhi and was approved by ethical committee.

Antimicrobial susceptibility testing

Antimicrobial susceptibility testing was performed by disc diffusion method for nalidixic acid (30 µg), ciprofloxacin (5 µg) and pefloxacin (5 µg) (Oxoid, UK) using Mueller Hinton agar (BD, Sparks, MD). The inhibition zone diameters were read using reflected light by two independent technologists and results were interpreted as per CLSI guidelines [20]. Discrepancies were resolved with discussion and if required retesting.

Minimum inhibitory concentration (MIC) for ciprofloxacin was established using Etest (BioMérieux, Marcy l'Etoile, France) on Mueller-Hinton agar (BD) following the manufacturer's instructions. Etest has been previously validated in our laboratory and results were comparable to that of the CLSI reference broth microdilution (BMD) method (unpublished data). The E-test results were rounded off to nearest log₂ dilution to compare with BMD reference MIC value, as suggested in previous studies (Deak et al., 2015; Humphries et al., 2019). Both MIC and disc diffusion were performed in parallel, using the same suspension, for each isolate. Controls included *Escherichia coli* ATCC 25922, *Pseudomonas aeruginosa* ATCC 27853, *Staphylococcus aureus* ATCC 29213. MIC of ciprofloxacin and zone diameters of nalidixic acid, ciprofloxacin and pefloxacin discs were within the established Quality Control range for the controls included in the study. The department also participates in the external quality assurance programme conducted by a national body. The breakpoints used in the study for ciprofloxacin MICs (µg ml⁻¹) were ≤0.06, 0.125–0.5

Table 1. Primer sequences used in the study

Genes	Primers for QRDR genes	DNA band size
<i>gyrA</i>	FP – 5'-TAC ACC GTC GCG TAC TTT ACG –3' RP – 5'-ACG CCA GGC AGC CGT TAA TC-3'	458bp
<i>gyrB</i>	FP – 5'- GCG CTG TCC GAA CTG TAC C– 3' RP – 5'- TGA TCA GCG TCG CCA CTT C –3'	169bp
<i>ParC</i>	FP – 5'-CTA TGC GAT GTC AGA GCT G-3' RP – 5'-CGC TTA ACA GCA GCT CGG-3'	265bp
<i>ParE</i>	FP– 5'-CCT TGT GGA AGG GGA TTC GG-3' RP – 5'-ATC GAT ACG GTA TAG CGG CG-3'	365bp
Gene	Primers of PMQR genes	Amplicon size
<i>qnrA</i>	FP:5'-ATT TCT CAC GCC AGG ATT TG-3' RP:5'-GAT CGG CAA AGG TTA GGT CA-3'	516bp
<i>qnrB</i>	FP: 5'-GAT CGT GAA AGC CAG AAA GG-3' RP: 5'-ACG ATG CCT GGT AGT TGT CC-3'	469bp
<i>qnrS</i>	FP: 5'-ACG ACA TTC GTC AAC TGC AA-3' RP: 5'-TAA ATT GGC ACC CTG TAG GC-3'	417bp
<i>aac(6')- ib-cr</i>	FP: 5'-TTG CGA TGC TCT ATG AGT GGC TA-3' RP: 5'-CTC GAA TGC CTG GCG TGT TT-3'	365bp
<i>qepA</i>	FP: 5'-GCA GGT CCA GCA GCG GGT AG-3' RP: 5'-CTT CCT GCC CGA GTA TCG TG-3'	1100bp

Source of primers–Eurofins, Bangalore, India.

and ≥ 1 for susceptible (S), intermediate (I) and resistance (R) respectively. The break points for S I R for ciprofloxacin (5 μ g), and nalidixic acid (30 μ g) disc were ≥ 31 , 21–30, ≤ 20 mm and ≥ 19 , 14–18, ≤ 13 mm respectively. Pefloxacin breakpoints for S and R were ≥ 24 and ≤ 23 mm respectively. No break point for intermediate category have been proposed for pefloxacin [21].

Characterization of fluoroquinolones resistance genes

The DNA from 180 *S. Typhi* isolates and the wild-type Ty2 reference strain of *S. Typhi* (GenBank accession number-AE014613) were extracted from a single colony after inoculation in 5 ml of Luria – Bertani broth and overnight incubation at 37 °C in shaker incubator. DNA extraction was performed using the Wizard genomic DNA purification kit (Promega,

Madison, WI, USA) as per the instructions. PCR was performed to study the FQ resistance mechanism. Primers used in the study (Table 1) were obtained from Eurofins, Bengaluru, India.

The PMQR genes screened were *qnrA*, *qnrB*, *qnrD*, *qnrS*, *aac(6')-Ib-cr*, and *qepA*. The quinolone resistance determining region (QRDR) of *gyrA*, *gyrB*, *parC*, and *parE* genes were amplified by PCR and mutations were confirmed by sequencing. Amplified PCR product (50 μ l) was run on 2% agarose gel and DNA was extracted from the gel by Qiagen kit (QIAGEN India Pvt. Ltd, New Delhi, India) and send to Macrogen (Korea) for sequencing. The alignment of sequence of the test DNA with the known sequence for *S. Typhi* Ty2 (controls) was performed using Finch Tv and Mega Blast software.

Data analysis

The disc diffusion results of the three surrogate discs (ciprofloxacin 5 μ g, nalidixic acid 30 μ g and pefloxacin 5 μ g) were evaluated to determine their performance to detect FQ resistance in *S. Typhi* isolates. Ciprofloxacin MICs were used as reference method for validating the performance of the surrogate disc and is based on the assumption that ciprofloxacin MICs (Using CLSI reference method) accurately differentiates FQ susceptible and FQ non-susceptible isolates. Infections with isolates showing DCS/ intermediate susceptibility to ciprofloxacin (MIC 0.125–0.5 μ g ml⁻¹) are associated with poor clinical outcome following treatment with FQ [16]. Hence in this study, DCS isolates were grouped together with the isolates showing resistance (MIC >0.5 μ g ml⁻¹) for further analysis. For each antimicrobial disc studied, scatterplots were generated using WHO NET 5.6 programme with zone diameters on x axis and ciprofloxacin MICs on y axis and discrepancy rates were calculated as per CLSI recommendations [22]. Very major error (VME; no of isolates false susceptible/ total ciprofloxacin non-susceptible isolates), major error (ME; no of isolates false non-susceptible/ total no of ciprofloxacin susceptible isolates), using the current *Salmonella* break-points for each surrogate disc, were calculated. Categorical agreement defined as a susceptible or non-susceptible that is consistent with ciprofloxacin MICs were also calculated. Discrepancy rates obtained were compared to the acceptable rates (Table 2) as per CLSI document M23-A4 [22].

RESULTS

Among the 180 isolates of *S. Typhi* included in the study, a fluoroquinolone resistance mechanism was identified in 146 isolates. Among them 84 isolates exhibited decreased ciprofloxacin susceptibility (DCS). Mutations in the DNA gyrase (*GyrA* and *GyrB*) alone or in combination with Topoisomerase IV (*ParC*) was the most common mechanism of resistance. PMQR was not detected. These isolates were phenotypically classified into three groups based on nalidixic acid and ciprofloxacin MICs and were distinct based on mutations in QRDR (Table 3).

Table 2. Criteria used to determine if antimicrobial susceptibility results are acceptable [22]

Error type	Results		Acceptable error rate	
	REF method	AST validated	N ≥ 30	N ≥ 100
Very major (VME)	R	S	0	< 3%, min 35 R isolates tested
Major (ME)	S	R	< 5%	< 3%
Minor (mE)	R	I	Major +minor < 10%	Major +minor < 7%

Very major error (VME; no of isolates false susceptible/ total ciprofloxacin non-susceptible isolates), major error (ME; no of isolates false non-susceptible/ total no of ciprofloxacin susceptible isolates), using the current *Salmonella* breakpoints. categorical agreement defined as a susceptible or non-susceptible that is consistent with ciprofloxacin MICs.

As shown in Table 3, Group 1 (N=68) included nalidixic acid susceptible (NAS) isolates. Within this group thirty-four of these isolates were wild-type (NAS, MIC range 1.5–1 µg ml⁻¹) with no mutations in the QRDR and ciprofloxacin MICs of 0.006–0.023 µg ml⁻¹. The remaining 34 isolates were also NAS (MIC range 4–12 µg ml⁻¹) with a single mutation in codon 464 of GyrB (Ser464 Phe, Ser464 Tyr or Ser464 Thr) and majority of these isolates (N=29) were non-susceptible, showing decreased susceptibility to ciprofloxacin (DCS) with MIC range 0.094–0.5 µg ml⁻¹ and for the remaining five isolates the ciprofloxacin MICs were raised (0.064 µg ml⁻¹) but within the susceptible range. The NAS isolates with GyrB mutation were thus distinct with MIC of nalidixic acid (3–12 µg ml⁻¹) and ciprofloxacin (0.064–0.5 µg ml⁻¹) being 3- and 12-fold higher respectively than the wild-type NAS isolates.

Group 2 (N=56) included nalidixic acid resistant (NAR) isolates with MIC range 96–512 µg ml⁻¹ and with a single mutation in codon 83 of GyrA (Ser83Phe, Ser 83Tyr) or single mutation in codon 87 of GyrA (Asp 87Asn, Asp87Gly). Among three isolates an additional mutation in ParC was detected. These isolates were non-susceptible to ciprofloxacin (CIP MIC range 0.094–0.5 µg ml⁻¹) except one isolate with single mutation in codon 87 of GyrA (Asp 87 Gly) with ciprofloxacin MIC of 0.064 µg ml⁻¹ (Table 3). The Group 1 (non-wild-type) and Group 2 isolates could be distinctly differentiated based on nalidixic acid susceptibility.

Group 3 (N=56) included nalidixic acid and ciprofloxacin resistant (NAR CIP^R) isolates and in majority of these isolates, double mutations in GyrA and a single ParC mutation (CIP MIC range 0.75–64 µg ml⁻¹) were detected. However, in this group, among three isolates (CIP MIC range 0.75–3 µg ml⁻¹) a single Ser83Phe mutation was detected.

Scatter plots correlating the MIC of ciprofloxacin vs nalidixic acid for 180 isolates are shown in Fig. 1. Although there is some heterogeneity among these 180 isolates, three Groups – 1 (wild and non-wild), 2 and 3, could be identified based on the MIC data for ciprofloxacin and nalidixic acid.

Effect of mutations on CIP MIC

Effect of various mutations on ciprofloxacin MICs for 180 isolates is shown in Fig. 2a. Ciprofloxacin MICs depicted a trimodal distribution, wild-type (no mutations, MIC ≤ 0.06 µg ml⁻¹) and ciprofloxacin resistant isolates (three mutations, MIC > 0.5 µg ml⁻¹) were identified distinctly and accurately by the breakpoints. However, isolates with a single mutation in GyrB or GyrA (MIC: 0.064–3 µg ml⁻¹) spilled over the wild-type isolates (MIC: 0.006–0.064 µg ml⁻¹) and resistant isolates (MIC > 0.5 µg ml⁻¹) respectively.

Table 3. Phenotypic characteristics and genetic mechanism of resistance to fluoroquinolones among *S. Typhi* (n=180)

Group	Phenotype	No. of isolates	Mutation in QRDR				Nalidixic acid MIC range µg ml ⁻¹	Ciprofloxacin MIC range µg ml ⁻¹
			GyrA	GyrB	ParC	ParE		
1	NAS (N=68)	34	Wild	Wild	Wild	Wild	1.5–3	0.006–0.023
		29	Wild	Ser464 Phe (N=13) Ser 464 Tyr (N=2) Ser 464 Thr (N=1) Ser 464 Tyr (N=13)	Wild	Wild	4–12	0.094–0.5
		5	Wild	Ser464 Phe (N=5)	Wild	Wild	3–12	0.064
2	NAR (N=56)	55	Ser83 Phe (N=24) Ser 83 Tyr (N=26) Asp 87 Asn (N=5)	Wild	Ser 80 Iso (N=1) Ser 80 Iso (N=2)	Wild	96–512	0.094–0.5
		1	Asp 87 Gly (N=1)	Wild	Wild	Wild	512	0.064
3	NAR CIP ^R (N=56)	53	Ser 83 Phe Asp 87 Asn	Wild	Ser 80 Iso	Wild -	512	2–64
		3	Ser 83 Phe (N=3)	Wild	Wild	Wild	512	0.75–3

CLSI breakpoints for susceptible, intermediate, and resistance, for ciprofloxacin 0.06, 0.125–0.5, >1 µg ml⁻¹ and nalidixic acid ≤16 µg ml⁻¹ and ≥32 µg ml⁻¹. NAS-Nalidixic acid susceptible, NAR, Nalidixic acid resistant; NAR CIP^R Nalidixic acid and ciprofloxacin resistant. PMQR was not detected.

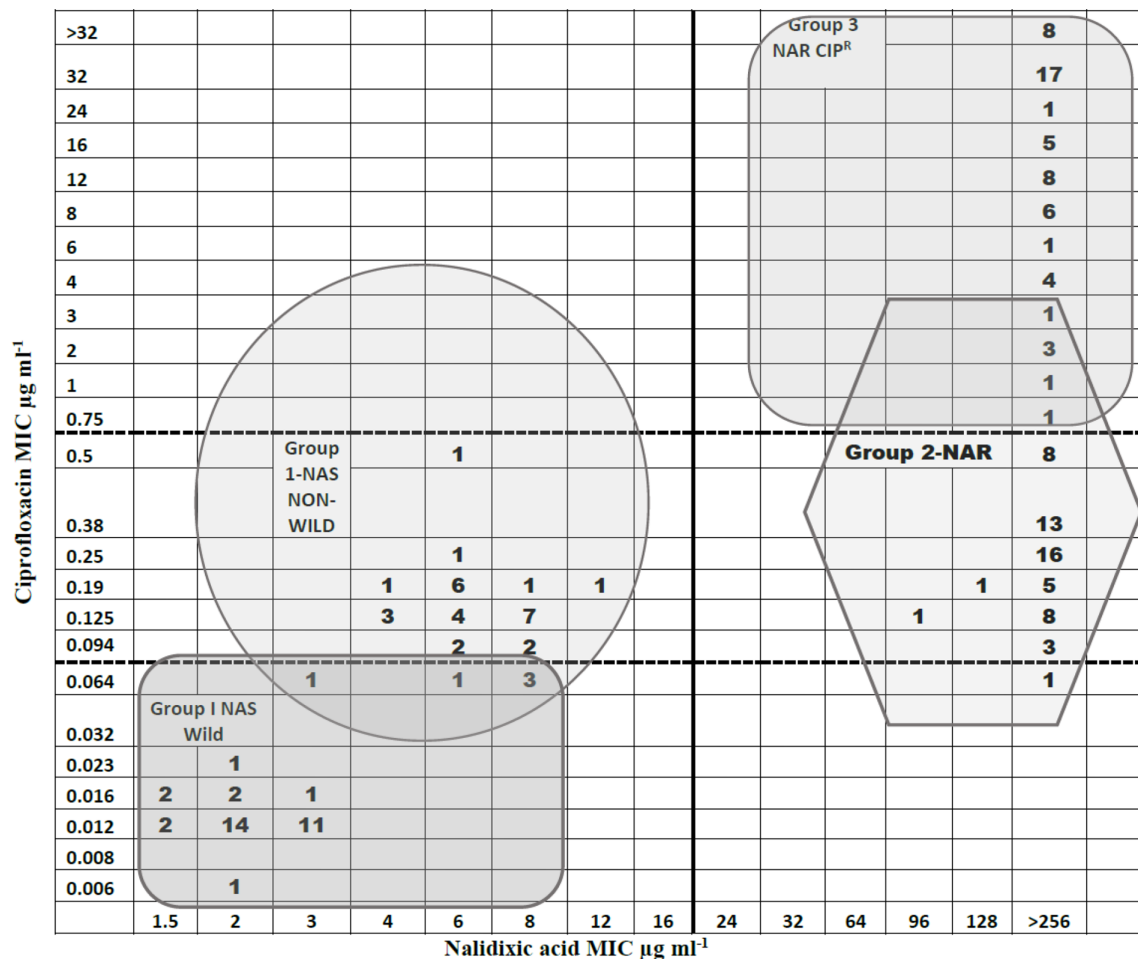


Fig. 1. Scatter plot: Nalidixic acid vs ciprofloxacin MIC ($\mu\text{g ml}^{-1}$) for 180 *S. Typhi*. Three Groups –1 (wild and non-wild), 2 and 3, could be identified based on the MIC data.

Validation of pefloxacin

To validate the performance of pefloxacin discs ($5\mu\text{g}$), for detection of isolates with altered susceptibility (MIC of ciprofloxacin $>0.06\mu\text{g ml}^{-1}$), results were compared with performance of nalidixic acid ($30\mu\text{g}$) and ciprofloxacin ($5\mu\text{g}$) discs. Disc diffusion results of the three drugs were compared with resistance mechanism and MIC of ciprofloxacin.

Performance of ciprofloxacin, nalidixic acid, and pefloxacin disc tests to detect isolates with various resistant mechanism are shown in Figs 2b–4 respectively. Ciprofloxacin disc ' $5\mu\text{g}$ ' could reliably detect (Fig. 2b) wild-type isolates using the break point of $\geq 31\text{ mm}$. Isolates with resistance mechanism (GyrA and GyrB) were non-susceptible to ciprofloxacin (Intermediate, range 20–30 mm or resistant zone diameter $\leq 19\text{ mm}$), however two isolates with GyrA mutation were susceptible (zone diameter 34 mm and 35 mm).

The nalidixic acid disc test could identify isolates without mutations (wild-type) as susceptible (zone diameter $\geq 19\text{ mm}$, range 22–38 mm) and isolates with single or double mutations in GyrA with or without ParC as nalidixic

acid resistant (zone diameter $\leq 13\text{ mm}$). For isolates with GyrB mutation, 12/34 isolates were non-susceptible to nalidixic acid (zone diameter range 12–18 mm) and 22 isolates were susceptible (zone diameter range 19–24 mm). Our results confirm that nalidixic acid disc test is only reliable to detect isolates with resistance mediated by mutations in GyrA with or without ParC mutations (Fig. 3).

The breakpoints proposed for detection of wild-type isolates using pefloxacin is $\geq 24\text{ mm}$ and could reliably detect and differentiate wild-type isolates in this study [23]. However as shown in Fig. 4, the low end of wild-type isolates and the high end of non-wild-type isolates (mutations in QRDR of *gyrA* and *gyrB* genes) coincided with an inhibition zone diameter of 24 mm, there was an overlap between isolates without and with FQ resistance mechanism.

The above data suggest that no single disc could reliably detect all isolates with altered/decreased susceptibility to ciprofloxacin mediated by mutations in GyrA/GyrB. To further evaluate the performance, scatterplots for each disc versus MICs of ciprofloxacin were constructed and

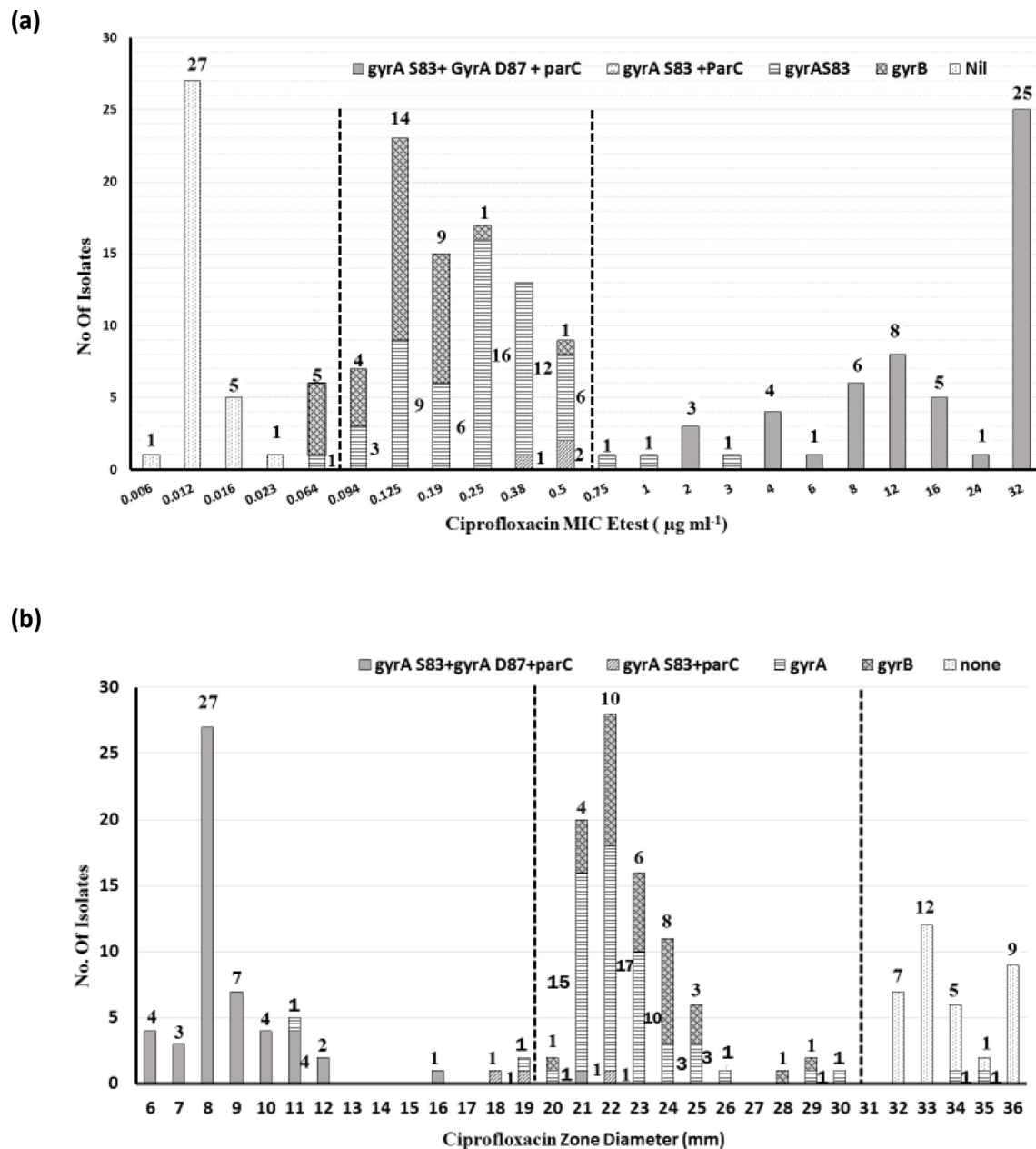


Fig. 2. (a) Effect of mutations on ciprofloxacin minimum inhibitory concentrations (MIC). CIP MICs depicted a trimodal distribution, wild-type and CIP^R isolates (three mutations in QRDR) were identified distinctly and accurately by the breakpoints. However, isolates with a single mutation in GyrB or GyrA (0.064–3 μg ml⁻¹) spilled over the wild-type isolates (0.006–0.064 μg ml⁻¹) and resistant isolates (MIC >0.5 μg ml⁻¹). (b) Ciprofloxacin inhibition zone diameter with mutations in QRDR of *S. Typhi* (n=180). Inhibition zone diameter of ciprofloxacin (5 μg) disc for *S. Typhi* 180 isolates with various fluoroquinolone resistance mechanism shown as different patterns. Breakpoints for Susceptible, Intermediate and Resistance are shown as broken lines.

are shown as in Fig. 5(a–c). The comparison was based on very major error, major error and categorical agreement (VME, ME and CA respectively) using susceptible and non-susceptible definitions as described earlier. The results are presented in Table 4).

The performance of nalidixic acid was poor with VME, ME and CA being 13.8, 2.6 and 90% respectively (Table 4).

VME were observed with 17 isolates with GyrB mutations which were susceptible to nalidixic acid (zone diameter range 19–22 mm) and non-susceptible to ciprofloxacin MICs (range 0.094–0.125 μg ml⁻¹). ME was observed in one isolate with GyrA D87 mutation, susceptible to ciprofloxacin (MIC 0.064 μg ml⁻¹) and resistant to nalidixic acid (zone diameter of 10 mm).

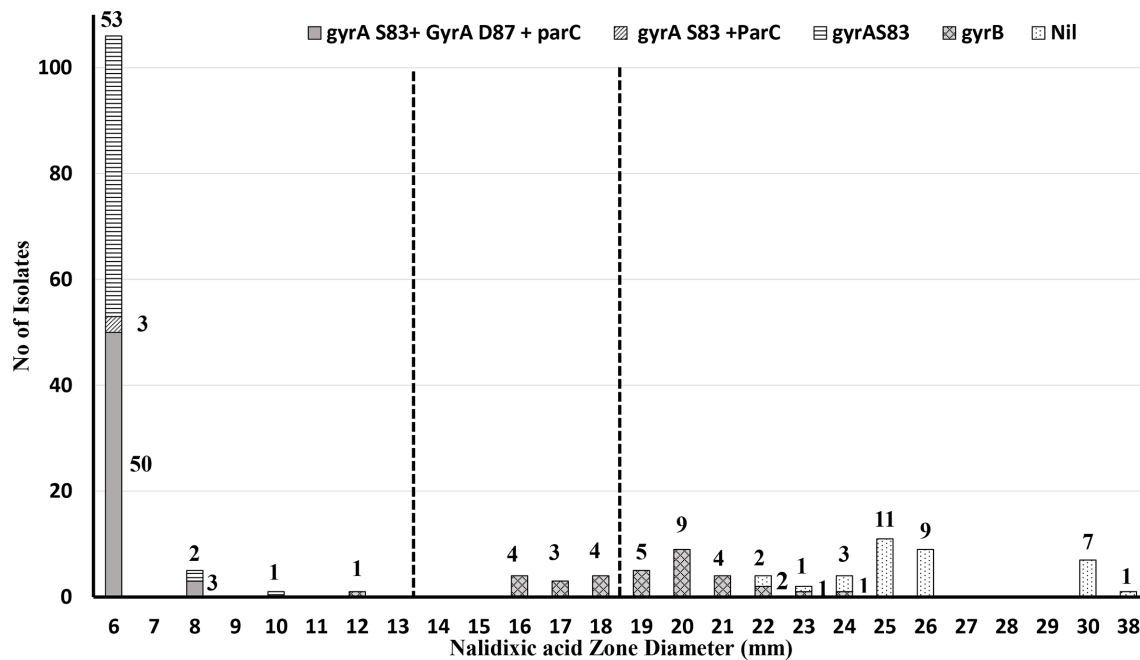


Fig. 3. Nalidixic acid inhibition zone diameter and Mutations in QRDR of *S. Typhi* ($n=180$). Inhibition zone diameter of nalidixic acid (30 μg) disc for *S. Typhi* isolates ($n=180$) with various fluoroquinolone resistance mechanism are shown as different patterns. Breakpoints for Susceptible, Intermediate and Resistance are shown as broken lines.

The performance of ciprofloxacin disc with reference method showed 0.7% VME, 12.5% ME and 96.7% CA. VME was observed with one isolate with single GyrA S83 mutation with decreased ciprofloxacin susceptibility (CIP MIC 0.38 $\mu\text{g ml}^{-1}$) and inhibition zone diameter within

susceptible range (34 mm). ME were observed among five isolates with GyrB mutation and their ciprofloxacin MICs were within the susceptible limit (0.064 $\mu\text{g ml}^{-1}$) and non-susceptible by disc test (zone diameter 22–29 mm).

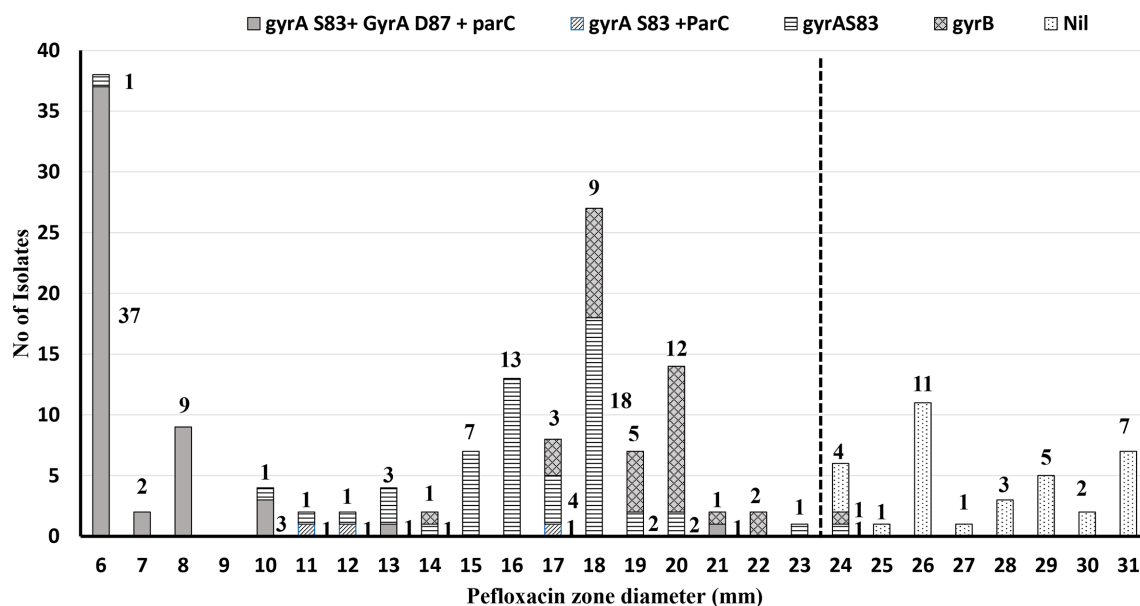


Fig. 4. Pefloxacin discs diffusion and mutations in QRDR of *S. Typhi* ($n=180$). Inhibition zone diameter of pefloxacin (5 μg) disc for *S. Typhi* 180 isolates with various fluoroquinolone resistance mechanism shown as different patterns. Breakpoints for Susceptible, intermediate and Resistance are shown as broken lines.

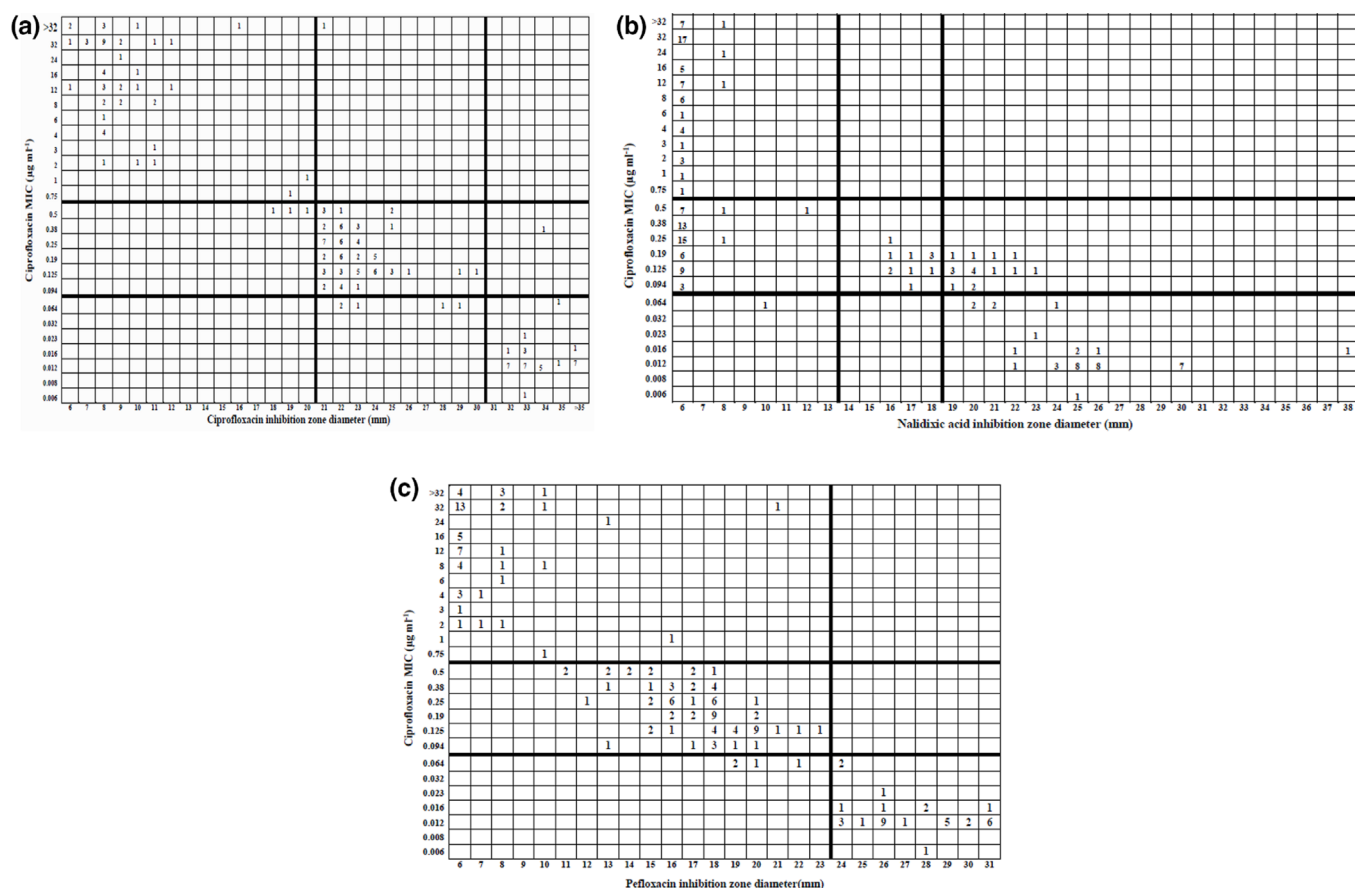


Fig. 5. (a) Scatter plot showing ciprofloxacin MIC versus ciprofloxacin zone diameter demonstrating ciprofloxacin MICs and disc correlations for 180 *S. Typhi* isolates. Ciprofloxacin breakpoints are $\leq 0.06 \mu\text{g ml}^{-1}$, susceptible and $\geq 1 \mu\text{g ml}^{-1}$ as resistant and ciprofloxacin zone diameter were interpreted as $\geq 31 \text{ mm}$ as susceptible, $21\text{--}30 \text{ mm}$ as intermediate and $\leq 20 \text{ mm}$ as resistant. (b) Scatter plot showing ciprofloxacin MIC versus nalidixic acid zone diameter. Scatter plot demonstrating ciprofloxacin MICs and nalidixic acid disc correlations for 180 *S. Typhi* isolates. Nalidixic acid breakpoints are $\geq 19 \text{ mm}$ as susceptible, $14\text{--}18 \text{ mm}$ as intermediate and $\leq 13 \text{ mm}$ as resistant. (c) Scatter plot of ciprofloxacin MIC versus Pefloxacin zone diameter demonstrating ciprofloxacin MICs and disc correlations for 180 *S. Typhi* isolates. Ciprofloxacin breakpoints are $\leq 0.06 \mu\text{g ml}^{-1}$, susceptible and $\geq 1 \mu\text{g ml}^{-1}$ as resistant and pefloxacin zone diameter were interpreted as $\geq 24 \text{ mm}$ as susceptible and $\leq 23 \text{ mm}$ as resistant [23].

Pefloxacin disc $5 \mu\text{g}$ performed better with 0% VME, 10% ME and 97.7% CA. ME were observed among four isolates with GyrB mutation, the ciprofloxacin MICs were within susceptible range ($0.064 \mu\text{g ml}^{-1}$) and non-susceptible by pefloxacin disc test (zone diameter 19–22 mm).

Irrespective of the screening disc, most of the discrepancies were found to be associated with isolates with GyrB mutation. We therefore constructed a scatterplot of 146 isolates, excluding isolates with GyrB mutation. The performance of all three surrogate discs improved with errors $< 3\%$ and VME,

Table 4. Comparison and performance of Ciprofloxacin, Nalidixic acid and Pefloxacin disc vs Ciprofloxacin MICs

S. Typhi	Ciprofloxacin disc $5 \mu\text{g}$			Nalidixic disc $30 \mu\text{g}$			Pefloxacin disc $5 \mu\text{g}$		
	VME%/N False S ^{**}	ME%/N False R ^{**}	CA%	VME%/N False S ^{**}	ME%/N False R ^{**}	CA%	VME%/N False S ^{**}	ME%/N False R ^{**}	CA%
N=180	0.71 (1)	12.5 (5)	96.7	12.1 (17)	2.56 (1)	90	0	10 (4)	97.7
N=146 (excluding GyrB mutation)	0.90 (1)	0	99.3	0	2.9 (1)	99.3	0	0	100

S. Typhi (n=180): *Based on 140 resistant isolates, **Based on 40 Sensitive isolates. Isolates excluding GyrB mutation (n=146): *Based on 111 resistant isolates, **Based on 35 Sensitive isolates. CA, categorical agreement (same 'Susceptible' or 'not Susceptible' results).

ME and CA agreement being 0, 2.9, 99.3%; 0.9, 0, 99.3 and 0, 0, 100% for nalidixic acid, ciprofloxacin and pefloxacin discs respectively. VME with ciprofloxacin and ME with nalidixic acid were due one isolate each with resistance mediated by GyrA mutation. These findings suggest that both nalidixic acid and ciprofloxacin discs fail to detect all isolates with GyrA or GyrB mutations and pefloxacin detected all isolates with GyrA mutations only.

The overall performance of ciprofloxacin and pefloxacin discs were comparable (Table 4). To further compare the performance of these two discs to reliably differentiate ciprofloxacin susceptible from the non-susceptible isolates (based on ciprofloxacin MIC, as shown in scatter plots), the percentage of isolates with inhibition zone diameter near the proposed breakpoint for each surrogate disc, arbitrarily chosen as 3 mm on either side of the breakpoint was calculated. Twelve percent of the isolates ($n=22$, Fig. 5a, c) included in the study had zone diameter within 3 mm of the breakpoints for both ciprofloxacin (28–33 mm) and pefloxacin (21–26 mm) suggesting that the performance of the discs was comparable.

DISCUSSION

The detection of low-level resistance or decreased susceptibility to ciprofloxacin (DCS) among *S. Typhi* has been a challenge for diagnostic laboratories. In endemic countries percentage of *S. Typhi* isolates with decreased ciprofloxacin susceptibility (DCS) have increased from 76.1% in 2004 to 97.0% in 2011 [10]. Most isolates with DCS when investigated were found to harbour single mutation in *gyrA* gene which encodes a subunit of DNA gyrase located in QRDR. Mutations in GyrA also confer high level resistance to non-fluorinated quinolone nalidixic acid and this has been typically described as classical quinolone resistance (NAR-DCS). Various investigators have reported that infections with *S. Typhi* showing NAR-DCS phenotype are associated with sub-optimal clinical response following FQ treatment [16, 17]. Based on these findings, nalidixic acid (NAL) resistance was proposed as the surrogate marker for FQ resistance by CLSI prior to 2012. However, NAL disc test is no longer considered reliable for detection of DCS with emergence of diverse FQ resistance mechanisms in *Salmonella* [10, 14, 23].

Numerous reports of non-classical quinolone resistance where nalidixic acid susceptible isolates with DCS (NAS-DCS) in both typhoid and non-typhoid *Salmonella* have emerged. The plasmid mediated quinolone resistance (PMQR) determinants such as *qnr* and *aac* (6')-*ib-cr* genes are associated with NAS-DCS phenotype with only modest increase in MIC of nalidixic acid (8–32 $\mu\text{g ml}^{-1}$) [18, 23]. PMQR has been commonly reported predominantly among non-typhoid *Salmonella* [18]. Mutations in GyrB also result in DCS (MIC 0.125–0.5 $\mu\text{g ml}^{-1}$) but retain susceptibility to NAL (MIC 2–16 $\mu\text{g ml}^{-1}$) [10, 14]. Mutations in GyrB have been reported among 1–11.6% of the *S. Typhi* in the France, United states and UK [14, 24, 25] and 9% from endemic countries like India [10]. These strains remain undetected due to widespread screening using nalidixic disc tests and limited investigations

undertaken to study mutations in GyrB. The impact of these resistance mechanisms on FQ treatment of various *Salmonella* infections is unknown as there have been no studies that document outcomes for such infections with FQs. There is a clear relationship between increasing ciprofloxacin MICs and outcome during fluoroquinolone treatment of typhoid fever. Infections with DCS isolates require higher concentrations of FQ to prevent *in vivo* selection of mutations associated with high levels FQ resistance. Moreover, under reporting of DCS by failure to detect isolates with NAS DCS phenotype may facilitate subsequent emergence of high level FQ resistance among *Salmonella*. Therefore, for clinical management, any increase in fluoroquinolone MICs irrespective of resistance mechanisms needs to be detected. A screening disc test to detect isolates with DCS especially in resource limited countries where ciprofloxacin MIC testing is generally not an option is urgently required.

In January 2012, CLSI revised ciprofloxacin MICs and zone diameter breakpoints specific for *Salmonella*. In view of the diagnostic challenges for detection of isolates with low level FQ resistance, CLSI in addition recommended the use of pefloxacin 5 μg disc for detection of these isolates. Various studies have validated the performance of pefloxacin disc to detect low level FQ resistance in both non-typhoidal and typhoidal *Salmonella*. Skov *et al.* [26] studied 146 non-typhoidal *Salmonella* isolates and the study by Deak *et al.* [23] includes 104 non-typhoidal and 32 typhoidal *Salmonella*. In typhoid endemic areas, Kapil *et al.* validated pefloxacin in 412 *S. Typhi* and *S. Paratyphi A* [27]. The predominant mechanism of resistance observed in these isolates was either mutations in GyrA or PMQR and none of the isolates demonstrated non-classical quinolone resistance (NAS-DCS) mediated by GyrB mutations [23, 26, 27]. Till date pefloxacin has not been validated for detection of low level FQ resistance in *S. Typhi* mediated by GyrB mutations. In this study from India where typhoid is endemic, we have evaluated performance of three surrogate discs (nalidixic acid, ciprofloxacin and pefloxacin) to identify a disc that most reliably identifies *Salmonella Typhi* isolates with a FQ resistance mechanism. These discs were evaluated against MICs of ciprofloxacin using susceptible or non-susceptible (I+R) MIC breakpoints. A diverse collection of isolates of *S. Typhi* collected between 2005–2014 and representing various phenotypes and diverse mechanisms of resistance were studied (Table 3). The resistance mechanisms were identified as mutations in GyrA, GyrB and ParC. PMQR was not detected in our isolates and has been very rarely reported among typhoidal *Salmonella* both in endemic and developed countries [28].

In the present study, performance of NAL disc (30 μg) was poor as it fails to detect isolates with DCS mediated by GyrB which are now emerging among *S. Typhi* in both endemic and developing countries [10, 24, 25]. However, NAL resistance is a good surrogate marker for detection of DCS mediated by mutations in the GyrA sub-unit of DNA gyrase and is in agreement with previous reports [23]. The other limitation of NAL disc reported in earlier studies is its inability to detect isolates with DCS mediated by PMQR determinants

Table 5. Characteristics of isolates with VME and ME errors with pefloxacin and ciprofloxacin surrogate disc test

ID no.	S. no	GyrA		GyrB	ParC	ParE	Disc zone diameter mm			MIC (Etest $\mu\text{g ml}^{-1}$)		Errors
		S83	D87				CIP 5 μg	NAL 30 μg	PEF 5 μg	CIP	NAL	
158	772	–	+	–	–	–	35	10	24	0.064	>256	None
44	147	+	–	–	–	–	34	6	18	0.38	>256	VME CIP
100	53	–	–	+	–	–	23	20	19	0.064	8	ME CIP and PEF
105	22	–	–	+	–	–	28	20	24	0.064	8	ME CIP
96	316	–	–	+	–	–	29	24	20	0.064	8	ME CIP and PEF
97	921	–	–	+	–	–	22	21	19	0.064	6	ME CIP and PEF
99	40	–	–	+	–	–	22	21	22	0.064	3	ME CIP and PEF

CIP, Ciprofloxacin; NAL, Nalidixic acid; PEF, Pefloxacin; MIC, Minimum inhibitory concentration; VME, Very major error; ME, Major error.

like *qnr* and *aac6-Ib-cr* reported predominantly among non-typhoidal *Salmonella* [18, 19]. In our study nalidixic acid disc test had a sensitivity of 87.9% (123/140) and a specificity of 97.5% (39/40) for detection of isolates with ciprofloxacin MIC of $>0.06 \mu\text{g ml}^{-1}$, however the sensitivity /specificity for pefloxacin (5 μg) and ciprofloxacin disc was 100%/ 90% and 99.2%/87.5% respectively. The performance of both ciprofloxacin (100%) and pefloxacin 33/34, 97%) discs was excellent with regard to differentiating isolates with GyrB mutations from wild-type as compared to nalidixic acid (22/34, 64.7%). The VME and ME were least with pefloxacin (0 and 10%) and ciprofloxacin (0.7 and 12.5%).

Table 4 shows the characteristics of the isolates with VME and ME errors with pefloxacin and ciprofloxacin discs. Isolate (772) with GyrA D57 mutation demonstrated zone diameters within susceptible range with both ciprofloxacin (zone diameter of 34 mm) and pefloxacin (zone diameter of 24 mm) and no error was assigned, as ciprofloxacin MICs were also within susceptible range near breakpoint ($0.064 \mu\text{g ml}^{-1}$). Isolate (147) with GyrA S83 mutation, and ciprofloxacin MIC of $0.38 \mu\text{g ml}^{-1}$ was correctly detected by pefloxacin disc (resistant, zone diameter of 18 mm) and a VME (false susceptible) observed with ciprofloxacin disc as the isolate was susceptible with a zone diameter of 34 mm on repeated testing. Both these isolates were detected by nalidixic acid disc test. ME were observed with both pefloxacin (10%) and ciprofloxacin (12.5%) disc and is higher than the typically accepted rate of $<3\%$. and may be skewed in the present study due to inclusion of few ($n=34$) wild-type isolates. All isolates with ME were associated with GyrB mutation and the ciprofloxacin MICs for these five isolates were within susceptible range (Table 5) and were factually false resistant by disc diffusion to ciprofloxacin ($n=5$, zone diameter range 18–26 mm) and pefloxacin disc ($n=4$, zone diameter range 19–22 mm). These finding suggest that different resistance mechanisms and mutations have a varied effect on the MICs of ciprofloxacin which may be raised several folds, compared to wild-type isolates, but remain within the susceptible range. Secondly the currently defined ciprofloxacin MIC breakpoint for susceptible ($\leq 0.06 \mu\text{g ml}^{-1}$) and non-susceptible

($\geq 0.125 \mu\text{g ml}^{-1}$, failed to detect all isolates with a resistance mechanism (Fig. 1b). These finding are in agreement with the reported MIC of 0.032 – $0.19 \mu\text{g ml}^{-1}$ and 0.03 – $0.05 \mu\text{g ml}^{-1}$ among isolates with GyrB mutation from France, USA and UK [14, 24, 25]. Further among the 460 isolates of *S. Typhi* studied by Parry *et al.*, it was observed that an MIC of $0.125 \mu\text{g ml}^{-1}$ to ciprofloxacin detected only 74.5% (341/460) of the *S. Typhi* strains with an identified fluoroquinolone resistance mutation in GyrA [29]. Studies on human and veterinary *Salmonella* have observed isolates with mutations in GyrA and ciprofloxacin MICs ($0.064 \mu\text{g ml}^{-1}$) within susceptible range [12, 23]. The clinical significance of these isolates needs to be investigated.

Our study confirms that none of the surrogate discs will identify all the isolates with a resistance mechanism. However, in countries where typhoid is endemic, detection of isolates with DCS is important to optimize therapy and monitor emerging mechanisms of resistance. Based on above data, both pefloxacin (5 μg) and ciprofloxacin (5 μg) discs can be used for screening classical and non-classical FQ resistance mediated by both GyrA (NAR DCS) and GyrB (NAS DCS) mutations respectively in *S. Typhi*. Mutation in GyrB is emerging as an important mechanism in *S. Typhi* and have been reported from both endemic and developing countries. These have been largely under-reported in endemic countries where CIP MICs are not routinely performed and till recently nalidixic acid discs were used for screening FQ resistance. Although CLSI and EUCAST have recommended use of pefloxacin for detection of FQ resistance and the only limitation of this disc is its inability of detecting PMQR mediated by *aac(6')-Ib-cr* gene, which is rare and has not been reported among typhoidal *Salmonella*. Based on the finding of isolation of NAS –DCS phenotype among *S. Typhi*, we propose screening for DCS using pefloxacin. Additional screening with nalidixic will help to differentiate between genetic mechanisms of resistance. Resistance to both or R R phenotype is indicative of GyrA mutation and S R phenotype is indicative of GyrB mutation.

This study had several limitations, firstly MIC of ciprofloxacin were determined by Etest and not by CLSI reference broth

micro dilution method. However, we found that MIC testing by two methods was comparable and this is further supported by a study by Deak *et al.*, where Etest was demonstrated to be a reliable option for testing ciprofloxacin for detection of low-level resistance in both typhoidal and non-typhoidal *Salmonella* [23]. The other limitation of this study was the lack of isolates with PMQR. However, this mechanism is rarely reported among *S. Typhi*. Another limitation of the study was that we tested only 35 isolates with no resistance mechanism hence the ME error rate may be skewed. The strength of this study is that this is the first report of validation of pefloxacin for detection of *S. Typhi* with non-classical FQ resistance mediated by GyrB.

To summarize and conclude, no single disc detects all mechanisms of resistance. This is the first report of validation of ciprofloxacin, nalidixic acid and pefloxacin discs for detection of low level FQ resistance among isolates of *S. Typhi* with both classical (NAR DCS) non-classical quinolone resistance (NAS DCS). Introduction of pefloxacin as a screening test will improve reporting of NAS-DCS phenotype which has till recently been under reported due to diagnostic challenges in endemic countries.

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Conflicts of interest

The authors declare that there are no conflicts of interest.

Ethical statement

This study was conducted in VMMC and Safdarjung Hospital, New Delhi and was approved by ethical committee.

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